Wrinkle-Assisted Nanofluidic Memristors for Tunable Ionic Memory and Neuromorphic Learning

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주름 기반 나노유체 멤리스터를 이용한 조절 가능한 이온 메모리 및 뉴로모픽 학습

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Abstract

Electronic memristors have accelerated neuromorphic research, yet their electron transport and complex fabrication limit biological fidelity and scalability. We present a wrinkle-based nanochannel array device (WNAD) formed by selectively preserved nanoscale wrinkles within a OSTEMER–OSTEMER bridge region, enabling simple, low-cost, and geometry-tunable ionic memory. By modulating nanochannel number and length, the WNAD exhibited strong memristive hysteresis and reproduced key synaptic behaviors including STP, LTP, PPF, and stable potentiation-depression. Higher-order processes such as learning, forgetting, and relearning further highlighted geometry-dependent memory consolidation. This work introduces a scalable, bio-inspired nanofluidic platform for next-generation ionic memory and neuromorphic computing.

Keywords: Micro-/Nanofluidics, Unconventional nanofabrication, Wrinkle-based nanochannels, Synaptic plasticity, Geometry-dependent tunable ionic memory

1. Introduction

Neuromorphic computing seeks alternatives to nanofluidic memristors limited by complex nanofabrication[1]. Iontronic devices mimic synaptic behavior via confined ion transport but typically require costly nanoscale structures. Here, we introduce a WNAD integrated into a PDMS/OSTEMER platform, where wrinkle-based nanochannel number and length are tuned by geometry. The device exhibited voltage-dependent ionic memristive and synaptic behaviors, offering a scalable and low-cost route to iontronic neuromorphic systems.

2. Results and Discussion

Figure 1 illustrates the selective formation of wrinkle-based nanochannels governed by the elastic modulus at the bonding interfaces. Figure 1A-B schematically shows the WNAD, in which wrinkle-based nanochannels were confined to the bridge channel connecting two mainchannels. The cross-sectional view highlights the contrasting behaviors of PDMS–OSTEMER and OSTEMER–OSTEMER interfaces, the low modulus of PDMS caused wrinkle collapse and suppressed nanochannel formation, whereas the higher rigidity of OSTEMER preserved the wrinkle morphology and enclosed nanochannels. SEM images (Figure 1C) confirm this modulus-dependent behavior, demonstrating that intact nanochannels existed only in the OSTEMER–OSTEMER bridge region, with therefore served as the active ionic transport pathway.

Figure 2 presents the geometry-dependent memristive behavior of the WNAD. Simulations result in Figure 2A-B show voltage-polarity-dependent ion dynamics, where forward bias induced ion accumulation within the nanochannels, while reverse bias lead to ion depletion. Figure 2C demonstrates that the hysteresis loop area increased systematically with the number of nanochannels, indicating enhanced ionic memory with increased parallel transport pathways. Figure 2D further reveals that the normalized hysteresis loop area scaled with both nanochannel number and length, confirming that the memristive response was governed by geometric control of ionic transport.

Figure 3 demonstrates geometry-dependent synaptic functions of the WNAD[2]. In Figure 3A, synaptic weight retention following 20 write pulses was evaluated using subsequent read pulses, revealing enhanced memory stability with increasing nanochannel number. Figure 3B shows paired-pulse facilitation as a function of nanochannel number and length, indicating that both parameters modulated short-term synaptic plasticity through ionic relaxation dynamics. Figure 3C presents potentiation and depression behaviors under sequential voltage pulses, while Figure 3D illustrates the reproducibility of these synaptic responses over ten consecutive potentiation-depression cycles, confirming stable and repeatable neuromorphic learning governed by device geometry.

Figure 4 illustrates learning-forgetting-relearning and associative learning behaviors in wrinkle-based nanochannels. Figure 4A shows synaptic weight evolution under consecutive write-read sequences, exhibiting learning, partial forgetting, and accelerated relearning dynamics analogous to biological synaptic adaptation. Figure 4B demonstrates geometry-dependent memory consolidation, where only the device with w = 75 μm and l = 50

μm underwent a transition to long-term potentiation after repeated training cycles, highlighting the critical role of nanochannel geometry in stabilizing long-term ionic memory.

3. Conclusions

In summary, we demonstrated wrinkle-based nanochannel array device (WNAD) that enables geometry-dependent ionic memristive and synaptic behaviors using a simple and scalable fabrication strategy. Selective formation of enclosed nanochannels was achieved by exploiting elastic modulus contrasts at PDMS–OSTEMER and OSTEMER–OSTEMER interfaces, confirming ionic transport exclusively to a well-defined bridge region. Combined simulations and experiments revealed voltage-dependent ion accumulation and depletion, with memristive hysteresis scaling systematically with nanochannel number and length, confirming that ionic memory is governed purely by geometry. The device further exhibited key synaptic functions, including short-term plasticity, paired-pulse facilitation, potentiation–depression cycles, and learning-forgetting-relearning behaviors. Notably, geometry-dependent memory consolidation enabled a transition to long-term potentiation only under optimized nanochannel configurations, highlighting the critical role of structural design in stabilizing ionic memory. These results establish WNADs as a low-cost, accessible, and scalable platform for iontronic memory elements and neuromorphic computing architectures.

Acknowledgments

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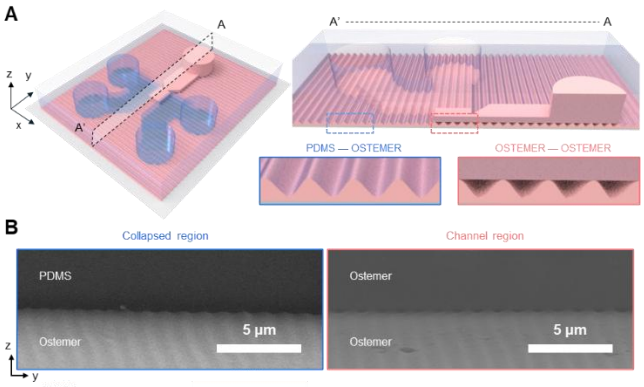


Fig. 1. Illustration of the WNAD and its characterization. (A) Schematic overview of the PDMS/OSTEMER hybrid chip integrated with channel and collapsed regions. (C) Cross-sectional SEM images of the channel and collapsed regions, confirming wrinkle-based nanochannel formation.

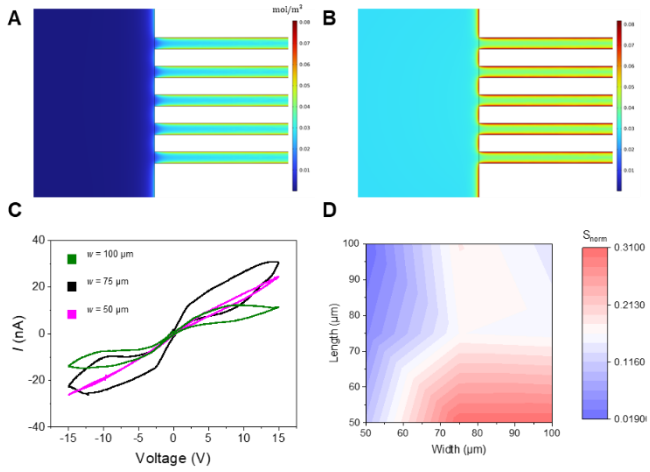


Fig. 2. Ion distributions along the wrinkle-based nanochannels and their memristive behavior. (A-B) Simulated cation concentration distribution near the junction under a forward/reverse bias, showing the accumulation/depletion dynamics. (C) Memristive hysteresis depending on the number of nanochannels. (D) Normalized hysteresis loop areas plotted with respect to the number and length of nanochannels.

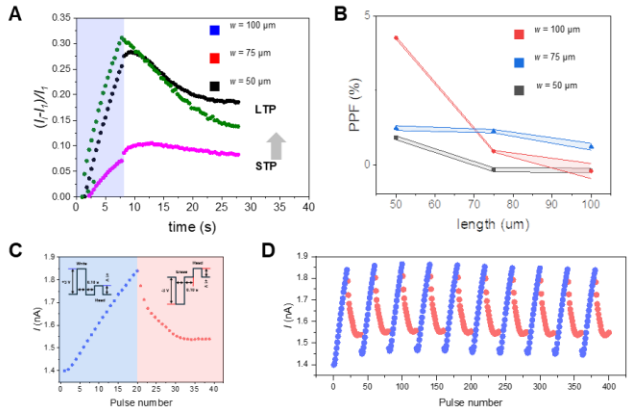
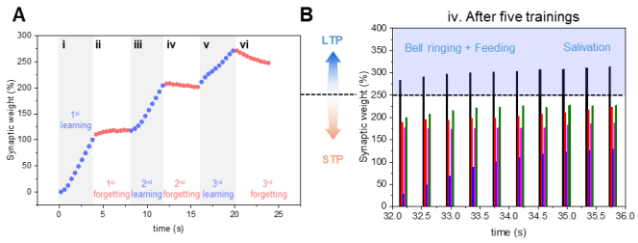


Fig. 3. Emulation of synaptic plasticity using the WNAD. (A) Synaptic weight modulation depending on the number of nanochannels. (B) Paired pulse facilitation (PPF) show dynamic Short-term plasticity (STP). (C) Cyclic potentiation-depression behavior under a stimulation protocol consisting of 20 write pulses followed by 20 read pulses.

Fig. 4. Application of the WNAD for emulating learning-



forgetting-relearning and associative learning. (A) Learning-forgetting-relearning behavior of the device reproduced consecutive write-read sequences. (B) Effects of the geometrical dimensions of the bridge channel on memory consolidation.